

**AMENDMENT NO. 2 APRIL 1999
TO
IS 1989 (PART 2) : 1986 SPECIFICATION FOR
LEATHER SAFETY BOOTS AND SHOES
PART 2 FOR HEAVY METAL INDUSTRIES**

(Fourth Revision)

(Page 10, clause 9, Title) — Substitute 'CONSTRUCTION' for 'MANUFACTURE'.

(Page 10, clause 9.2, line 4) — Substitute '3 mm' for '3 cm'.

(Page 10, clause 9.4) — Substitute the following for the existing text:

'The counter shall be turned over so as to form a jugloop as shown in Fig. 1. The distance between top of quarter to the top of jugloop shall be 20 mm. The joint of the quarters shall be reinforced by stitching a wide strip of leather at the back. A strengthening leather piece (see 8.17) shall be pasted and stitched at the centre of the inside quarter.'

(Page 22, Appendix B) — Substitute the following for the existing text:

**' APPENDIX B
(Clauses 12.1 and 18.1)**

DETERMINATION OF IMPACT VALUE

B-1 GENERAL

The impact test is carried out for determining the performance of safety boots reinforced with steel toe cap to withstand a blow of 14 kgf.m.

B-1.1 Test Specimen

B-1.1.1 The test shall be made on the toe of the finished safety boots/shoes, sampled from each size of a lot.

B-1.1.2 Pattern of Boot

The test shall apply only to safety boots and shoes fitted with protective steel toe caps.

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B-1.1.3 Preparation of Test Specimen

Prepare test specimen from the forepart of the footwear by cutting off the toe end not more than 30 mm from the rear edge of the toe cap. The upper and lining shall not be separated.

B-2 APPARATUS

B-2.1 The testing apparatus shall be such that a 27.0 ± 0.2 kg load can be allowed to fall freely on vertical guides from various predetermined heights to strike a cylindrical mild steel plunger, 38 mm in diameter and 145 mm in length. Before carrying out any test, the test apparatus shall be checked for its ability to fall freely without any restriction up to the striking bar at the base of the apparatus. The plunger shall be freely supported in a vertical guide, and shall have attached to its lower end a horizontal mild steel bar 155 mm long, 38 mm wide and 10 mm thick. The bar (strike plate) shall be capable of resting across the toe-cap with its front edge in line with the point of toe-cap. The upper end of the plunger shall have a mild steel plate of $63 \text{ mm} \times 63 \text{ mm}$, screwed to it, and can be replaced when damaged. The above dimensions shall have a tolerance of $\pm 1 \text{ mm}$.

B-2.1.1 The base of the machine shall be solidly constructed of hard wood of not less than 75 mm thick. To this a metal block (50 mm thick) shall be bolted to support the steel plate on which the boot rests.

B-2.1.2 A capsule as illustrated in Fig. 5 suitable for measuring clearance inside the boot/shoe at the moment of impact shall be used. The capsule consists of a small duraluminium plunger which is depressed on impact into a brass cylinder. The metal clip shall be fastened around a rubber tubing which encases the cylinder and by tightening and loosening the clip the amount of friction between the cylinder and plunger can be varied. The cylinder wall shall be slotted to enable the clip to compress the cylinder on to the plunger. The clip shall be adjusted so that the movement between the two parts takes place at a load of less than 14.0 N, but no movement takes place at a load of 8.0 N.

B-2.1.3 Stabilizing Fork

The details of a suitable stabilizing fork have been given in Fig. 6A and 6B.

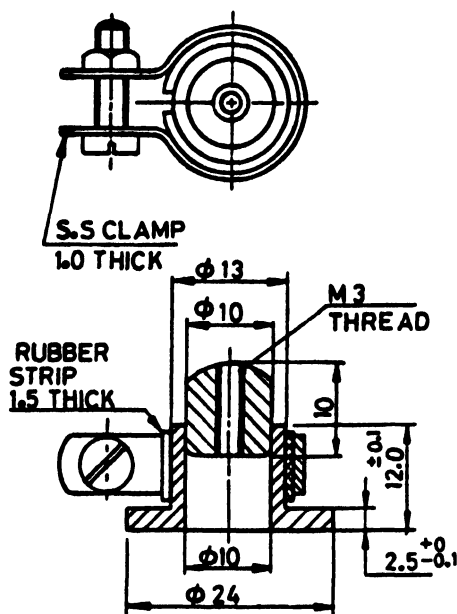
B-3 PROCEDURE

B-3.1 Positioning

Place the capsule in the slot of the stabilization fork and position it close to the rear end of the slot. Then place the test specimen on the clamping base so that the fork tips touch the fore end of its inner lining. The details of a suitable

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stabilizing fork have been given in Fig. 6A and 6B. Fix the stabilizing fork by means of the clamping screw to rest on the insole. Now adjust both the clamping screw and the adjusting screw so that the stabilizing fork becomes more or less parallel to the base plate and the teat piece is damped firmly (see Fig. 8).



All dimensions in millimetres.

FIG. 5 CAPSULE FOR MEASURING THE CLEARANCE AT THE
MOMENT OF IMPACT

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Now, place the clamping device of the impact testing equipment with the securely clamped test piece under the vertical plunger of the performance testing apparatus (see Fig. 7 and Fig. 8).

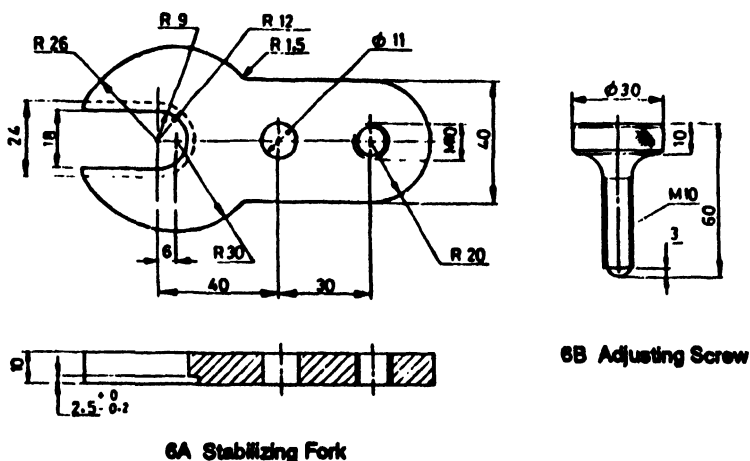
Position the vertical plunger about 1 mm inside the rear edge of the steel toe-cap and under the striking bar so that the capsule is behind the centre line of the bar. The bar shall test on the boot lengthwise roughly at right angles to the length of the test specimen.

B-3.2 Impact

Adjust the load to a height of 508 ± 5 mm above the top of the vertical plunger (see Fig. 8) and allow it to fall freely and strike the plunger. The impact should be of 14 kgf.m.

B-3.3 Measurement

Remove the capsule by unscrewing the adjusting screw (see Fig. 6B) and the clamping screw and measure the length from the top of the duralumin plunger to the base of the capsule by means of a slide caliper fitted with a vernier, capable of measuring the value to the nearest 0.1 mm. This value is the clearance inside the boot/shoe at the moment of maximum depression.'



All dimensions in millimetres.

FIG. 6 ATTACHMENT FOR FIXING THE CAPSULE

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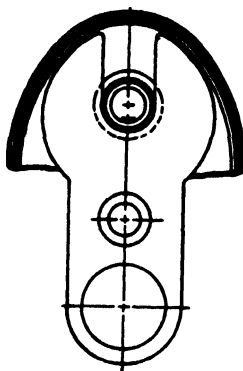


FIG. 7 PLACEMENT OF THE SAMPLE FOR DETERMINATION OF IMPACT VALUE

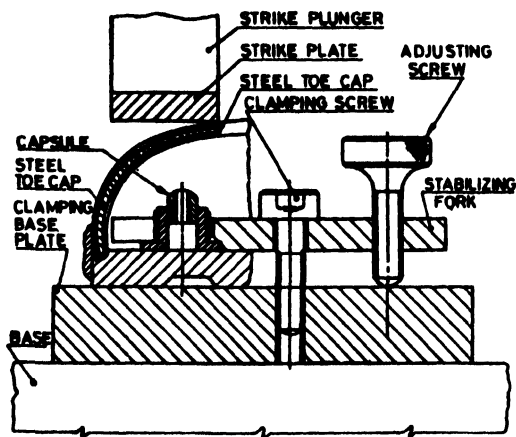


FIG. 8 POSITIONING AND CLAMPLING OF THE FOREPORT
OF THE SAFETY FOOTWEAR

(CHD 19)